

Portfolio Daemon — Project Overview

Project: Portfolio Daemon

Technology: Python, asyncio, asyncssh, httpx, watchdog

Location: /home/dominic/Documents/portfolio-daemon

Report #: 1/20

Overview

Background daemon managing multiple backend services with SSH tunnels and health monitoring.

Purpose

Manages local backend services and SSH tunnels for remote development.

Status

Active development. See report 16 (Known Issues) and report 19 (Improvements).

Key Features

• Auto-start backend services • SSH tunnel management • Health monitoring via / health • File-watch restart • PM2 integration

Entry Points

```
python main.py (daemon)
```

Detailed Architecture

Project Structure

```
daemon.py (main loop), config.py, services/ (per-service config), monitor.py (health checks), s
```

Technology Stack

Layer	Technology	Purpose
Backend Framework	Python	REST/API server
Database	SQLAlchemy + SQLite	Data persistence
Auth	JWT	Authentication & authorization
Frontend	None	User interface
Deployment	Render.com + GitHub Pages	Hosting & CI/CD

Module Breakdown

See project structure above.

Data Flow

1. Request enters via HTTP (REST, GraphQL, or WebSocket)
2. Authentication middleware validates JWT token (if required)
3. Router dispatches to appropriate handler
4. Handler processes business logic with database queries
5. Response formatted and returned to client
6. Frontend renders response in appropriate view

Entry Points

- **Server:** `python daemon.py`
- **Port:** 8000 (default)
- **Health check:** `GET /health`

Configuration

Environment Variables

Variable	Description	Default
DATABASE_URL	Database connection string	sqlite:///data.db
SECRET_KEY	JWT signing secret	(required)
CORS_ORIGINS	Allowed CORS origins	*
DEBUG	Debug mode flag	false

Deployment

Requirements

- Python 3.10+
- pip dependencies (see requirements.txt)
- SQLite (or PostgreSQL for production)

Startup Commands

```
# Development
python daemon.py --reload

# Production
python daemon.py --host 0.0.0.0 --port 8000
```

Related Reports

- Report 02: Architecture Analysis (detailed component diagram)
- Report 05: Tech Stack & Dependencies (version-specific)
- Report 13: Deployment Guide (Docker, CI/CD)